

TECHNICAL SPECIFICATION

Section 26 33 53 — Uninterruptible Power Supply (UPS) System

Project:	Dhanbad Tier-III Data Centre — Building A, Phase 1
Client:	NorthStar Data Infrastructure Pvt. Ltd.
Spec Section:	26 33 53 (CSI MasterFormat)
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Facility Class:	Uptime Institute Tier III (Concurrently Maintainable)

1. General

1.1 This section specifies the design, performance, and installation requirements for the double-conversion online Uninterruptible Power Supply (UPS) system serving the critical IT load of Data Hall 1. The system shall provide continuous, conditioned power and ride-through capability during utility disturbances until standby generation assumes load.

1.2 Referenced Standards: IEC 62040-1 (Safety), IEC 62040-3 (Performance/EMC), UL 1778, ANSI/TIA-942 (Data Center Infrastructure), NFPA 70 (NEC), IEEE 519 (Harmonic Control).

2. UPS System Performance Requirements

Parameter	Required Value	Test/Verification Method
Rated Capacity	800 kVA / 720 kW	Nameplate + factory load bank test
Topology	Double-Conversion On-Line (VFI)	IEC 62040-3 classification
Input Voltage	415 V AC, 3-Phase + N, $\pm 10\%$	Site commissioning test
Output Voltage	415 V AC, 3-Phase + N, $\pm 1\%$	Site commissioning test
Frequency	50 Hz $\pm 0.5\%$	Site commissioning test
Rated Efficiency (Online Mode)	$\geq 96.0\%$ at 100% load	Factory acceptance test (FAT) report
Output Power Factor	0.9 (capacity de-rated) or better	FAT report
Input Current THD	$\leq 5\%$ at full load	FAT report / IEEE 519 compliance
Battery Autonomy	≥ 10 minutes at full rated load	Battery discharge test
System Redundancy	N+1 (minimum 2 UPS modules)	Design drawing review
Overload Capability	125% for 10 min; 150% for 1 min	FAT report
Operating Temperature Range	0°C to 40°C	Manufacturer datasheet
Enclosure Protection Rating	IP20 (indoor, restricted access)	Manufacturer datasheet
Acoustic Noise Level	≤ 65 dBA at 1 m	Manufacturer datasheet
Communication / Monitoring	SNMP v3 + Modbus TCP, dry contacts	Integration test with BMS

Parameter	Required Value	Test/Verification Method
Parallel Capability	Up to 4 units, hot-swappable	Design drawing review

3. Battery System

Parameter	Required Value
Battery Type	Valve-Regulated Lead-Acid (VRLA), sealed
Design Life	>= 10 years at 25°C ambient
Monitoring	Cell-level voltage & temperature monitoring required
Battery Room Ventilation	Per IEEE 1635 / NFPA 1 hydrogen dilution requirements

4. Compliance & Documentation Requirements

- 4.1 Manufacturer shall submit factory acceptance test (FAT) reports prior to shipment.
- 4.2 All submittals shall reference this specification section and identify any deviations explicitly.
- 4.3 Equipment shall carry UL 1778 listing; documentation shall be included in the submittal package.
- 4.4 Non-conforming submittals will be returned for revision and re-submission prior to procurement approval.

End of Section 26 33 53. This document is a synthetic sample created for hackathon software testing purposes and does not describe a real facility.